REFEREE AID - VEHICLE COMBAT

Vehicle Combat for the purpose of this aid is defined as one vehicle targeting and shooting at another vehicle. If a vehicle is targeting infantry or exposed positions (or when firing machine guns), it is recommended to use the **Heavy Weapons & Explosives** or **Ranged Combat Referee Aids** as appropriate.

Vehicle Combat initiated → Draw initiative cards. Lowest card goes first. → Participant banks two fast actions, or one fast and one slow action. → Determine all To-Hit Modifiers. The attack is a slow action.

Movement or fast action(s) taken (often done by a driver).

Free Actions

Note: Aiming with mortars and howitzers is a slow action. Otherwise aiming is a fast action. Gunners of heavy weapons must always aim - there are no quick shots allowed.

Examples pg. 65:

Range S/M/L/E	-/-1/-2/-3
Player Specialties	+1
Target Hex Terrain	-1/-2
Target Prone (not same hex)	-1
Target moving	-1
Firing from moving vehicle	-2
Firing from elevation	+1
Light/Dim/Dark	-/-1/-2
Rain/Wind/Snow	-1
Dense Smoke	-3
Target in Full Cover	-3
Small Target (or a person)	-2
Artillery Correction	+1

Attacks on Vehicles:
Small Arms: Small Arms firing at vehicles gain a +2 modifier. Heavy Weapons do not receive this bonus, and it doesn't apply to bicycles nor motorcycles.
Moving vehicle: If the shooter is firing from a moving vehicle, they suffer a -2 modifier to the attack. Stabilizers negate this modifier.
Target movement: If the target vehicle has moved since the shooter's last turn, the attack is rolled with a -1 modifier.

Vehicle Movement:
Movement: Vehicles can move in combat up to their combat speed for on-road and off-road driving respectively, by using a fast action. Only paved hexes count as on-road. You can end a driving movement facing any direction you wish. This movement is known as the safe distance the vehicle can move.
Driving Roll: If you make a driving roll after travelling a safe distance on your turn, you can roll your driving skill to move your vehicles combat speed *plus* one hex for each of the number of successes achieved beyond the first. Failure on this driving roll means your vehicle gets stuck. You can push this roll, but failure means a loss of one reliability step. If this breaks the vehicle (0 Reliability), the vehicle breaks down after the safe distance, but before any extra movement is completed by the driving roll.
Getting Unstuck: Use a slow action to make a driving roll, modified by the terrain type. Success unsticks the vehicle. If you fail this roll, the vehicle becomes bogged down. To get unstuck from being bogged down, you must spend a stretch of time and make a stamina roll to get the vehicle unstuck. Other players can help, and if you use another vehicle as large as the stuck one it will provide a +2 modifier.

Ammo Dice can only be used with weapons that have a RoF. If a weapon has no RoF or only a single shot is fired, no ammo dice are used. pg. 66.

Add ammo dice in accordance with weapon ROF if relevant → Ranged Attack - Roll AGI and Heavy Weapons dice, plus any ammo dice used → Hit?

When firing mortars and howitzers, roll INT + Heavy Weapons instead.

Target Missed (End of participant's turn)

Push?

Target Missed (End of participant's turn)

Ramming:
In order to ram a vehicle or person, you must make an opposed skill check. The rammer must make a driving roll, as does the vehicle being hit. If a person on foot is targeted, they must roll mobility instead. If you hit, you inflict damage equal to the rear armor level of your vehicle, plus one for each you roll.
Ramming vehicles: When ramming a vehicle, the attacking vehicle suffers damage equal to the target's rear armor. The target is always hit on the side facing the attacker; the attacker always takes the damage to its front.
Ramming people: The crit threshold for ramming people is 2. The attacking vehicle suffers no damage.

Vehicle Facing Diagram / Angle of Attack

Additional Equipment Rules:
Smoke Launchers: If a vehicle has smoke launchers, they can be fired using a slow action. This fills all hexes surrounding the occupied hex by dense smoke.
Stabilizers: Vehicles with stabilizers don't suffer the -2 penalty modifier when firing on the move.
Fire Control Systems (FCS): Vehicles with FCS don't suffer the -2 penalty modifier when firing on the move. In addition, all ranged attacks are calculated as if at short range.
Vehicle Night Vision: Weapons equipped with Infrared (IR) sights have the negative penalty modifier removed for short range attacks made in darkness, or medium range attacks with IR spotlights turned on.
Vehicle Thermal Sights: Weapons with Thermal Sights have the negative penalty modifier removed at all ranges when firing in darkness, rain, fog, or snow.

Explosive ammunition: Please reference explosion deviation rules when using explosive types of ammunition. pg. 71.

Player may reroll and at their own discretion.

Reroll AG, HW, and any ammo dice that aren't s, s, or s.

Weapon jams can be cleared with a successful Ranged Combat or Heavy Weapons roll. pg. 66.

Mishap: A single : -1 Rel from weapon. 2 or more s: Weapon jam

Weapon base DMG. Add +1 for each extra Add +1 for each extra

*Check angle of attack**

**Enter the diagram here if switching from the HW&E guide.*

If armor level is 2 steps higher than the weapon's base damage, the attack is fully deflected. pg. 82

Apply weapon's armor modifier if there is one.

Armor?

Horses:
Movement: When riding horses, you can double its movement in the same way as vehicles, but roll mobility instead of driving.
Ranged Combat: Firing 1-handed while mounted gives a -2 penalty. Firing 2-handed gives a -3 penalty. The Rider specialty negates these penalties.
Close Combat: Attacking from horseback is possible with one-handed weapons. Attacks against the rider suffer a -1 penalty.
Damage: Horses have 5 hit capacity. Critical hits kill them. Incapacitated horses must be saved by a medical roll within a shift, or be put down. If a horse is hit, it's rider must roll a mobility skill check or be thrown off, suffering 1 point of damage.

ATGMs:
ATGMs strike their target the following round on the shooter's turn. They also use STR+HW to hit, unless fired from a tripod/vehicle mount, in which case they use AGI+HW.
Wire-Guided: Roll your To-Hit dice on the round after firing to see if you guided the missile in. If you become incapacitated/suppressed, you lose line of sight to the target, or you move, the shot automatically misses.
Fire-and-Forget: Roll when firing the missile. If you roll a , the missile will hit the target unless it moves beyond extreme range.

Penetration Rules:
If not all **direct damage** is negated by the target's armor, the shot penetrates. When this happens, roll 1d10 on the Penetration table below. The shot will destroy components (move up the component list) as it continues to move through the vehicle until it runs out of energy (damage). Otherwise if the shot doesn't penetrate, roll 1D10 on the No Penetration table.
Penetration: If a part listed doesn't exist or was already damaged, move up one step on the list until you reach a present and intact component.
No Penetration: Apply the full damage of the attack to the component hit. If a component doesn't exist or has already been destroyed, then the attack does no damage.

COMPONENT DAMAGE

D10	PENETRATION	NO PENETRATION
1	Fuel	Track/wheel*
2	Engine	Weapon
3	Suspension	Fire control system
4	Ammunition	Antenna
5	Cargo	External stores
6	Driver	External stores
7	Passenger	Exposed passenger
8	Gunner	Exposed passenger
9	Commander	Ricochet
10	Radio	Ricochet

*Re-roll for front and rear attacks.

If armor is penetrated by an attack, roll a base D6 die. On a the armor level is decreased by 1. Armor may be repaired with a Tech roll and a shift of work. Each repairs 1 level of armor.

Reduce damage by armor value. → Penetration? → Roll accordingly on the Component Damage Chart (see Penetration Rules) → Settle any damage to crew/passengers and applicable components

Remember to include any blast damage (from explosive penetrations) done to the crew! Information on recovering from damage and criticals can be found starting on pg. 73.

Blast Damage:
Many vehicle weapons also have an explosive effect that causes blast damage to those inside the blast radius. Please reference blast damage on pg. 68.

Additional Attack Information
Reloading: Reloading vehicle weapons usually takes a slow action. Having a loader allows the weapon to be fired every round.
Bailing Out: Anytime a vehicle is penetrated, all occupants must make a CUF roll. Failure means they must immediately spend a slow action exiting the vehicle.
Called Shots: You can make called shots (which never penetrate) on external components of a vehicle with a -2 modifier. The small arms modifier of +2 when firing at vehicles negates this. You can only hit the tracks or wheels when making a called shot from the sides.
Disabling Vehicles: Hits to the Engine, Fuel Tank, Suspension, Tracks, or Wheels of a vehicle can render the vehicle unable to move until it is repaired. A hit to the fuel tank or the ammunition on a vehicle can render the vehicle permanently destroyed and it cannot be repaired (it can still be scavenged for parts, however).
Explosive Hits: If a weapon with Blast Power penetrates the vehicle, all occupants suffer the effects from the explosion as well as any direct hits. The blast damage is not decreased by the armor.
Targeting a hex: When targeting a hex, ignore the target hex terrain type when calculating the to-hit dice being used. Extra successes beyond the first have no effect.
Ambush: You cannot ambush from a moving vehicle, but you can waylay enemies if you spend a stretch in preparation. pg. 61.

TWILIGHT: 2000

Component List - Hits and Effects:
Fuel: A penetration of 1 or more will cause the fuel tank to leak. The maximum fuel capacity is reduced by half until repaired, and the fuel tank will immediately drain to that level if it has fuel in excess of 50% in it. Any excess damage beyond 1 will carry on to other components.
If penetrated by an explosive attack, roll for the blast power as normal. If there are any successes, the fuel catches fire. All crew are exposed to fire with intensity C, and must make a CUF roll or bail out on their very next turn. A fuel fire destroys the vehicle beyond repair, and the fire will burn for a stretch.
Engine: Each point of penetration damage will reduce the reliability rating of the vehicle by 1. If reduced to zero, the vehicle becomes inoperable. Any excess damage will *not* transfer to other components, being stopped by the engine block.
Ammunition: The main armament's ammo has been hit. If 1 point of penetration is done, the ammunition is halved. If 2 points are done, all ammunition is lost. Excess penetration damage will carry on to the next component.
Ammunition for heavy weapons also has a risk of exploding. 1 point of penetration damage carries a 50% chance of exploding (1D3). 2 points of penetration damage will always cause the ammo to explode. If the ammunition explodes, it causes an explosion with the blast power of the ammunition exploding, raised one step (up to an A). The vehicle is also destroyed beyond repair.
Cargo: The internal cargo hold is struck, and a random piece of cargo suffers damage (pg. 91 for effects of damage on gear). If the item is destroyed, any additional damage transfers to other components.
Driver: The driver is hit, and suffers damage normally. Any excess damage transfers to another component. Any other occupant of the vehicle can become driver with a fast action.
Passenger: A random passenger is hit, and suffers damage normally. Any excess damage transfers to another component, including other passengers if present.
Gunner: The gunner is hit, and suffers damage normally. Any excess damage transfers to another component.
Commander: The commander is hit, and suffers damage normally. Any excess damage transfers to another component.
Radio: Each point of damage reduces the reliability rating of the radio by 1. Normally radios only have a reliability rating of 1. If reduced to 0, the radio becomes inoperable until it is repaired. Any excess damage transfers to another component.
Track/Wheel: Can only be hit by side attacks. For each point of damage in excess of half of the side armor of the vehicle (rounded up and then modified by the armor modifier), the reliability rating of the vehicle is reduced by 1. If reduced to 0, the vehicle becomes inoperable. If the attack's damage doesn't exceed half of the side armor level, then the attack does no damage.
Weapon: A random weapon on the vehicle has its reliability rating reduced by 1. If reduced to 0, the weapon becomes inoperable until its repaired. Tank guns are armored, so damage to them is reduced by half of the front armor of the vehicle (rounded up, then modified by armor modifier).
Fire Control System (FCS): If the FCS suffers 1 point of damage or more, it is rendered inoperable.
Antenna: If the antenna suffers 1 point of damage or more, it is rendered inoperable.
External Stores: A random piece of external cargo is hit. For damage effects on gear, please see pg. 91.
Exposed Passenger: A random exposed passenger is hit (or partially exposed), and suffers damage normally.
Ricochet: The attack bounces off in a random direction. Anyone on foot in the hex must make a CUF roll.